JIANGHUI SUN

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EDUCATION

University of Illinois Urbana-Champaign | Master of Computer Science | Aug 2026 – Dec 2027 (Expected)

Xi'an Jiaotong-Liverpool University | B.Sc. Information and Computing Science | Sep 2022 – Jul 2026

GPA: 3.8/4.0

RESEARCH & INDUSTRY EXPERIENCE

Research Assistant | UIUC SSAIL | Present

- Analyzing prefill/decode-disaggregated LLM serving for multi-turn agent workloads to identify compute bottlenecks and latency trade-offs across repeated inference cycles.
- Designing workload-level evaluations of prefill/decode placement, throughput, end-to-end latency, and GPU utilization to inform serving optimization; measurements in progress.

Research Assistant | Xi'an Jiaotong-Liverpool University | Nov 2025 – Jan 2026

Fine-grained Emoji Understanding via DINOv2–Qwen-VL

- Built a fine-grained emoji-understanding pipeline integrating DINOv2 visual representations with Qwen-VL language generation to capture subtle expression cues.
- Designed a feature-alignment adapter connecting DINOv2 embeddings to Qwen-VL vision inputs, enabling expression-conditioned descriptions without retraining the full language model.

Research Assistant | Xi'an Jiaotong-Liverpool University | Sep 2025 – Nov 2025

EmoSound: Emotion-Aware Audio Accompaniment | BICS 2025, Paper #52

- Developed a cross-attention soft-prompt adaptation of frozen CLIP for 1,613 emojis across 18 emotion classes; achieved 82.44% Top-1 accuracy and 82.31% macro F1, improving accuracy by 60.91 percentage points over vanilla CLIP.
- Built a VLM/LLM-agent audio-matching workflow with Top-5 retrieval from a 287-clip library; in a 10-person study, improved perceived expressiveness from 2.7/5 to 4.3/5 versus text-only labels.

ML Algorithm Engineering Intern | RocKontrol Technology Group | Jun 2024 – Aug 2024

- Benchmarked 4 PyTorch forecasting architectures (RNN, LSTM, GRU, Transformer) using 40K+ air-quality, emissions, and weather records and 24-hour observation windows.
- Delivered 6-, 12-, and 24-hour forecasts for PM2.5, O3, and AQI; selected LSTM after achieving ~12% lower RMSE than the RNN baseline.

SELECTED RESEARCH PROJECTS

Knowledge Organization for Agent Memory Evolution | UIUC CS546 | In Progress

- Developed a controlled evaluation design comparing flat and semantic memory organizations under matched knowledge and feedback budgets to isolate memory-update effects.
- Specified reinforcement-learning-based tests across 3 planned benchmarks (STALE, AgentStream, MemoryAgentBench), measuring stale-memory correction and valid-knowledge retention; experiments ongoing.

Cross-modal Audio Generation for Visually Impaired Users | Jun 2025 – Sep 2025

- Engineered a training-free CLIP/Qwen/FAISS emoji-to-audio retrieval pipeline using Top-5 image matching; organized 1,691 emoji images spanning 18 emotion categories.
- Curated a 420-clip audio library (279 collected, 141 AI-generated) and integrated AudioLDM2 music with AudioMiniMax speech synthesis into a working accessibility demo.

PUBLICATION

Sun, J., Shi, H., Yin, Z., Mo, W., Ding, H., Hu, Y., Yang, X., & Yan, Y. “EmoSound: A Multimodal AI Agent Framework for Emotion-Aware Audio Accompaniment of Emoticons.” BICS 2025 conference program, Paper #52.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, SQL

AI/ML: PyTorch, Transformers, scikit-learn, CLIP, LoRA/PEFT, multimodal learning, RAG

Tools: FastAPI, Docker, Git/GitHub, LangChain, Jupyter